

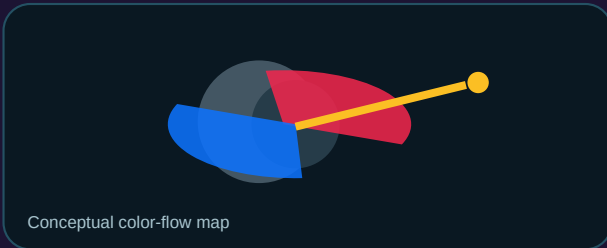
COLOR DOPPLER

VS

SPECTRAL DOPPLER

Fetal Follow-Up Echocardiography Cheat Sheet

COLOR = MAP



Where is blood flowing?

SPECTRAL = MEASUREMENT



How is blood moving?

THE CENTRAL IDEA

Color Doppler overlays a **spatial map** of mean Doppler shift on the 2D image. Spectral Doppler places velocity information on a **graph over time**. They answer different questions, produce different stored evidence, and support different coding elements.

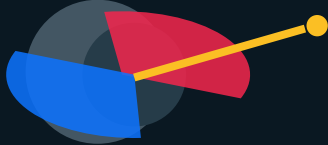
COLOR ALONE DOES NOT CREATE A SPECTRAL WAVEFORM

Direction on color depends on the displayed map and the transducer reference. Red and blue do not mean arterial/venous or oxygenated/deoxygenated blood.

One signal. Two representations.

Both begin with Doppler frequency shift. The display and clinical question are different.

COLOR DOPPLER



Conceptual color-flow map

A map inside the anatomy

- Encodes the sign and magnitude of mean Doppler shift within each color pixel.
- Localizes flow, jets, shunts, regurgitation, turbulence, and aliasing.
- Preserves the spatial relationship between flow and cardiac structures.
- Fast for surveying a region; less precise for exact peak velocity.

Primary question

Where is flow, and what direction/pattern is visible?

SPECTRAL DOPPLER



Velocity

Time ->

A waveform across time

- Displays the distribution of Doppler shifts from a sample volume or CW beam.
- Measures peak velocity, timing, direction relative to the baseline, and waveform shape.
- Supports inflow/outflow assessment, valve stenosis evaluation, rhythm analysis, and mechanical AV interval.
- Requires deliberate beam placement, scale, sweep speed, and waveform capture.

Primary question

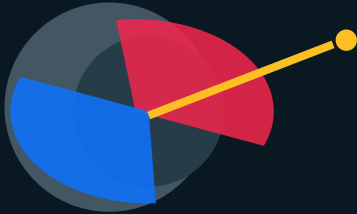
How fast, in which direction, and with what timing/pattern?

They are complementary, not interchangeable

- Color helps place the spectral sample gate in the correct jet, vessel, inflow, or outflow.
- Spectral Doppler then quantifies the selected signal across time.
- A complete clinical answer often uses 2D anatomy + color localization + spectral characterization.

Read the map before you trust the color.

Color is a directional overlay, not a diagnosis. Interpretation depends on scale, gain, box size, frame rate, and the displayed color bar.



Conceptual color-flow map

WHAT COLOR ADDS

- Spatial location of flow within the 2D plane
- Direction relative to the transducer and color map
- Jet origin, width, extension, and relation to anatomy
- Mosaic/aliasing pattern that can flag high velocity or disturbed flow
- Rapid survey of valves, septa, arches, and venous pathways

Typical fetal echo targets

SEPTA

VSD shunt; foramen ovale/interatrial flow

VALVES

Mitral/tricuspid regurgitation; semilunar flow

OUTFLOWS

LVOT/RVOT turbulence or abnormal direction

ARCHES

Aortic and ductal arch direction/patency clues

VEINS

Pulmonary/systemic venous connections when indicated

SHUNTS

Ductus arteriosus and other abnormal pathways

COLOR PITFALLS THAT CHANGE THE MAP

- Scale too low: aliasing may appear where flow is normal.
- Scale too high: low-velocity flow may disappear.
- Gain too high: blooming can overstate a jet or fill a defect.
- Large color box: lower frame rate and temporal resolution.
- Poor insonation angle: weak or absent signal despite real flow.
- Color reversal: read the map bar; never assume red = artery.

The waveform is the measurement record.

Spectral Doppler converts the selected Doppler signal into velocity distribution versus time.



INFLOW

Mitral and tricuspid

E- and A-wave morphology, fusion, timing, and direction

OUTFLOW

Aortic and pulmonary

Peak systolic velocity, contour, duration, and stenosis characterization

VENOUS / RHYTHM

DV and paired signals

Ductus venosus pattern; atrial-ventricular timing and mechanical AV interval

Acquisition controls that protect the measurement

- **Sample placement**
PW gate in the intended vessel/jet; CW samples the full beam path.
- **Beam alignment**
Velocity is angle-dependent. Align as parallel to flow as feasible.
- **Scale + baseline**
Avoid clipping and display forward/reverse components clearly.
- **Sweep speed**
Show enough cycles while preserving event timing and morphology.
- **Wall filter + gain**
Retain low-velocity information without excessive noise or spectral fill-in.
- **Freeze + trace**
Store representative waveforms and document measurements used clinically.

Angle note: For quantitative velocity, an unfavorable beam-flow angle can underestimate the true velocity. Color helps locate the target; it does not correct spectral angle error.

Choose the modality by the question.

Direction can be shown by both modalities, but color localizes it in anatomy while spectral displays it relative to a baseline over time.

CLINICAL QUESTION	COLOR	SPECTRAL
Where is flow occurring?	YES - spatial map	Not primarily
Which direction is flow going?	YES - map-dependent	YES - vs baseline
Is there turbulence or aliasing?	YES - rapid survey	Can characterize signal
Is there regurgitation?	Detects and localizes jet	Velocity/timing profile
What is the peak velocity?	Not the primary tool	YES
What is the pattern over time?	No waveform	YES
Can I characterize a gradient?	No	YES - velocity-based
Can I assess mechanical AV interval?	No	YES - paired signals

Fast selection rule

Need to **find or localize** the flow? Start with **COLOR**. Need to **quantify or time** the flow? Acquire **SPECTRAL**.

The best fetal echocardiographic answer frequently uses both.

Same anatomy. Different evidence.

These examples show how color localizes a finding and spectral Doppler adds velocity, timing, or waveform characterization when clinically indicated.

1 Fetal ventricular septal defect (VSD)

COLOR DOPPLER

Shows flow crossing the ventricular septum, identifies the jet origin, and demonstrates direction relative to the displayed map.

SPECTRAL DOPPLER

Can characterize direction, velocity, and timing of the shunt signal if the information will change interpretation or management.

2 Aortic stenosis

COLOR DOPPLER

Shows a narrowed color-flow column and turbulent or aliased flow across the aortic valve and into the ascending aorta.

SPECTRAL DOPPLER

Measures peak systolic velocity and waveform contour. Serial measurements can help characterize progression in context.

3 Tricuspid regurgitation

COLOR DOPPLER

Shows the regurgitant jet entering the right atrium and localizes its origin, direction, and spatial extent.

SPECTRAL DOPPLER

Displays the regurgitant velocity profile and timing when clinically indicated. Interpretation depends on signal quality and alignment.

Do not force a measurement. Spectral interrogation should answer a clinical question. Color helps identify the correct target, while an aligned, well-optimized spectral tracing supports quantitative interpretation.

The stored evidence must match the code.

This page is an educational workflow aid for follow-up fetal echocardiography. Verify current CPT guidance, payer edits, and local policy.

CPT 93325

Color-flow mapping

- Color mapping was actually performed.
- Representative color images/clips are stored.
- Findings describe direction, jets, shunting, regurgitation, or mapped flow as applicable.
- Add-on code: do not report as a solitary service.

CPT 76828

Follow-up spectral Doppler

- Spectral interrogation was actually performed.
- Representative PW/CW waveforms are stored.
- The report documents sites, waveform findings, and relevant measurements.
- Color images alone do not support this spectral Doppler code.

NO SPECTRAL WAVEFORM = NO 76828

COLOR ALONE DOES NOT SUPPORT 76828

Five-second workflow

- 1 MAP** Use color to localize flow and identify the target.
- 2 SAMPLE** Place PW gate or align CW through the target.
- 3 MEASURE** Optimize scale, baseline, sweep, angle, and tracing.
- 4 STORE** Archive representative color clips and spectral waveforms.
- 5 REPORT** Document what was performed and what it showed.

SELECTED SOURCES

1. Moon-Grady AJ, et al. ASE Guidelines and Recommendations for Performance of the Fetal Echocardiogram. JASE. 2023;36:679-723. 2. Carvalho JS, et al. ISUOG Practice Guidelines (updated): fetal cardiac screening. Ultrasound Obstet Gynecol. 2023;61:788-803. 3. ISUOG Practice Guidelines (updated): use of Doppler velocimetry in obstetrics. Ultrasound Obstet Gynecol. 2021;58:331-339. 4. Society for Maternal-Fetal Medicine. CPT Medical Coding for Maternal-Fetal Medicine, 2026 ed. 5. OpenMFM: Billing for Follow-up Fetal Echocardiography. openmfm.org/decks/billing-followup-fetal-echocardiography/

Educational aid. Coding must reflect the examination actually performed and documented.